Fictional Worked Example: Preserving Evidence from a Suspected Compromised Laptop

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Companion to	`docs/evidence-handling-guide.md` and `templates/chain-of-custody-template.md`

> **Everything in this example is fictional.** The college, people, device, serial numbers, seal numbers and hash values are invented for training. IP addresses come from private and documentation-only ranges (RFC 1918 and RFC 5737). No real device was examined, and no data was collected from any person's device. Hash values are illustrative strings in valid SHA-256 format; they do not correspond to real files.

1. Scenario

Meridian College of Applied Computing (fictional) runs a small IT Security team. At **10:22 IST on Wednesday 14 October 2026**, the college firewall raised an alert: a laptop in Research Lab 2 (`10.20.4.37`) was repeatedly connecting to an external address on a threat-intelligence blocklist (`203.0.113.45`).

The laptop is a college-owned Windows 11 Pro machine used by a research student. The IT Security Officer authorised a response, and a lab assistant was sent as the **first responder**. An external forensic examiner was called in to capture memory and image the disk.

2. People involved (all fictional)

Name	Role in this incident
R. Kale	Lab Assistant: first responder; author of the action log
A. Desai	IT Security Officer: incident lead; network isolation; evidence store custodian
Dr M. Rao	Head of IT: authorised the response
S. Iyer	External forensic examiner: memory capture, power-down, imaging
N. Shah	Research student: regular user of the laptop

3. Evidence register

Item	Description	Created or seized	Seal
E1	Laptop, Windows 11 Pro, "Fictional Model LX-14", S/N `LX14-DEMO-0007`, asset tag `MCAC-IT-LT-0412`	Seized	S-000316, later S-000341
E2	65 W USB-C charger belonging to E1	Seized	S-000317
E3	32 GB blue USB flash drive labelled "DEMO", found in E1's left USB-A port	Seized	S-000318
E4	Memory image of E1 (`memdump_E1.raw`), on the examiner's sterile drive	Created by examiner	N/A (digital)

Item	Description	Created or seized	Seal
E5	Scene photographs P-01 to P-10 (JPEG), taken on college work phone `MCAC-PH-07`	Created by responder	N/A (digital)
E6	Forensic disk image of E1	Created by examiner	N/A (digital)
E7	PDF scan of R. Kale's notebook pages 1–3 (action log)	Created by responder	Original notebook held by A. Desai

4. Observation and photo log

All times are by the **reference clock** (work phone on network time), IST (UTC+05:30).

Photo	Time	What it shows
P-01	10:40	Wide view of Desk 4: laptop open, charger and Ethernet connected, user's bag nearby (not touched)
P-02	10:40	Screen: desktop with Outlook open; Word not open
P-03	10:40	Left side: Ethernet connected; USB drive (E3) in the USB-A port
P-04	10:40	Right side: charger (E2) connected; charging LED on
P-05	10:40	Lid sticker showing asset tag `MCAC-IT-LT-0412` (photographed without moving the laptop)
P-06	10:41	Laptop clock (10:38) next to the reference phone (10:41)
P-07	10:46	Network icon showing "No internet" after network-side isolation
P-08	10:48	Close-up of USB drive E3, still inserted
P-09	12:24	Underside serial label, photographed after power-off, at the examiner's request
P-10	12:26	Sealed bags E1, E2 and E3 with seal numbers visible

5. Power-state decision record

Question	Answer
Power state on arrival	**ON**, unlocked, logged in as N. Shah, on charger
Any visible destruction (wiping, encryption in progress, ransom note)?	**No**
Encryption	BitLocker **enabled** (per A. Desai). Recovery key held in the college directory service. **Key not recorded.**
Decision (guide, Step 3)	**Keep powered. Do not shut down, log out or close the lid. Isolate network-side. Call examiner for memory capture.**
Reason	Volatile evidence (running processes, network connections, keys) would be lost at power-off, and BitLocker would lock the disk. No active destruction justified cutting power.
Decided by and time	R. Kale, confirmed by A. Desai by phone, 10:43

6. Contemporaneous action log (transcribed from notebook pages 1–3)

Clock note: At 10:41 the laptop clock showed **10:38**, while the reference showed **10:41**. The laptop clock is **3 minutes slow**, so add 3 minutes to any laptop timestamp to get real time. N. Shah's "about 10:15" was read from the laptop clock, so it is about 10:18 real time. That fits the firewall alert at 10:22.

Time (IST)	Who	Action	Reason	Result / observation
10:31	R. Kale	Received call from A. Desai about the firewall alert for `10.20.4.37`	Assigned as first responder	Collected kit; left for Lab 2
10:38	R. Kale	Arrived at Lab 2, Desk 4. Asked N. Shah to step away and not touch the laptop.	Secure the scene	User complied. "Restricted" sign placed on lab door.
10:39	R. Kale	Recorded user statement (**reported, not verified**)	Capture actions taken before arrival	N. Shah: opened email attachment `Scholarship_Form.docm` at about 10:15 (laptop clock); a blue window flashed; closed Word at about 10:20. Did nothing else.
10:40	R. Kale	Photos P-01 to P-05	Record before touching	See photo log
10:41	R. Kale	Clock comparison, photo P-06	Timeline alignment	Laptop 3 min slow
10:42	R. Kale	Assessed power state **without contact**	Guide Step 3	ON, unlocked, no visible destruction
10:43	R. Kale	Phoned A. Desai with power decision and encryption query	Specialist confirmation	Decision confirmed. BitLocker on; key location noted; key not written.
10:45	A. Desai (remote)	Disabled the switch port serving Desk 4	Network isolation without touching the laptop	Reported by phone
10:46	R. Kale	Observed network icon, photo P-07	Confirm isolation	"No internet". Ethernet cable left in place. Wi-Fi already off (icon); not touched.
~~10:84~~ 10:48 *(RK, 10:49)*	R. Kale	Noted USB drive E3 in left port, photo P-08	Minimise handling	**Left in place** for the examiner
10:55	R. Kale	Laptop locked automatically (idle timeout)	—	**Not touched.** Lock screen visible; laptop still powered.
11:05	R. Kale	Remained with laptop; no further contact	Preserve scene	—
11:40	R. Kale	Examiner S. Iyer arrived; briefed from notes and photos	Hand-over	—
11:42	R. Kale → S. Iyer	Custody transfer of E1, E2, E3 (in place, powered on)	Memory capture and power-down	Logged on CoC forms; A. Desai witnessed
11:48	S. Iyer	Asked N. Shah to unlock the laptop with their own password, witnessed by A. Desai	Required for memory capture under authorisation	Unlock recorded as a **change to system state**

Time (IST)	Who	Action	Reason	Result / observation
11:50–12:14	S. Iyer	Memory capture to the examiner's sterile external drive	Preserve volatile data	E4 created; SHA-256 recorded (Section 7)
12:18	S. Iyer	**Forced power-off** (held power button)	Avoid "Shut down", which writes to disk	Laptop off at 12:18
12:20	S. Iyer	Removed E3, bagged, seal S-000318	Protect original	—
12:24	S. Iyer	Photo P-09 of underside serial label	Record identifier	S/N `LX14-DEMO-0007`
12:25	S. Iyer	E1 in anti-static bag, seal S-000316; E2 seal S-000317; photo P-10	Protect original	Signed across seals
12:30	S. Iyer → A. Desai	Transfer of E1, E2, E3 to IT Security locker L-3	Secure storage pending imaging	Seals intact
13:05	R. Kale	Copied P-01 to P-10 from phone to the restricted evidence share; created SHA-256 manifest	Integrity of responder-created files	E5 manifest hash recorded
13:20	R. Kale	Scanned notebook pages 1–3 to PDF; hashed it	Preserve the log	E7 hash recorded. Original notebook given to A. Desai at 13:25.

The 10:48 entry shows the correction rule: the wrong time was struck through with a single line, initialled and timed. It was not erased.

7. Hash record (SHA-256)

Item	What was hashed	SHA-256 (illustrative)	By	Date / time (IST)
E4	`memdump_E1.raw`	`5664605743e29305853e00f40f940df3eae3d7f56a76dc1341fd2c5ad3abfd3b`	S. Iyer	2026-10-14 12:15
E5	`E5_photos_manifest.sha256` (lists each photo's hash)	`385d2805620faaee64b1a767c1ae8351f6881a01743c65a76c2b592c8bd9dd02`	R. Kale	2026-10-14 13:05
E7	`E7_action_log_scan.pdf`	`5267840f9abc604c327bbfb0260619e245cdc658f19e1fee7ec71d31011ba822`	R. Kale	2026-10-14 13:20
E6	Disk image of E1 (at acquisition)	`9ba11bd5ffc4816352da6ade8c9cb9bafdeba47ecdb2ba72d097774a3fc7611e`	S. Iyer	2026-10-15 11:02
E6	Working copy, verification	`9ba11bd5ffc4816352da6ade8c9cb9bafdeba47ecdb2ba72d097774a3fc7611e` (**match**)	S. Iyer	2026-10-15 12:30

No hashing tool was ever run **on E1 while it was live**. E1's disk was hashed only as part of imaging (E6), by the examiner, through a write blocker. A printed copy of this table is held by A. Desai, separately from the digital files.

8. Completed chain-of-custody form: item E1

A. Case information

Field	Entry
Case / incident reference	MCAC-IR-2026-017
Organisation / department	Meridian College of Applied Computing (fictional), Dept. of IT, Research Lab 2
Incident lead	A. Desai, IT Security Officer, ext. 4410
Authorised by	Dr M. Rao, Head of IT
Date and time authorised	2026-10-14 10:30 IST (UTC+05:30)

B. Evidence item description

Field	Entry
Evidence item number	E1
Item type and description	Laptop computer, Windows 11 Pro
Make / model	Fictional Model LX-14
Serial number	LX14-DEMO-0007 (underside label, photo P-09)
Asset tag / other identifiers	MCAC-IT-LT-0412 (lid sticker, photo P-05)
Colour, condition, visible damage	Grey; minor scratch on lid; no other damage
Accessories packaged with item	None (charger is a separate item, E2)
Power state when found	ON, unlocked, on charger
Power state when packaged	OFF: forced power-off by S. Iyer at 12:18 IST
Encryption status and recovery-key location	BitLocker enabled (per A. Desai). Key held in college directory service. Key not recorded.
Bag / seal number(s)	S-000316 (14 Oct); S-000341 (15 Oct, after imaging)

C. Collection

Field	Entry
Collected by	R. Kale, Lab Assistant
Signature	*R. Kale (signed)*
Date and time collected	2026-10-14 10:38 IST: control taken in place; not moved
Location collected	Research Lab 2, Desk 4
Photograph references	P-01 to P-10 (E5)
Notes	User reported opening an attachment and closing Word before arrival. Auto-locked 10:55; unlocked by user at 11:48 at examiner's request, witnessed by A. Desai.

D. Transfer log

#	Date and time (TZ)	Released by	Received by	Purpose	Seal no., intact?	Hash verified?	Location after transfer
1	2026-10-14 11:42 IST	R. Kale *(signed)*	S. Iyer *(signed)*	Memory capture and power-down	N/A (powered on, not yet bagged)	N/A	Research Lab 2, Desk 4
2	2026-10-14 12:30 IST	S. Iyer *(signed)*	A. Desai *(signed)*	Secure storage pending imaging	S-000316, Y	N/A	IT Security locker L-3
3	2026-10-15 09:30 IST	A. Desai *(signed)*	S. Iyer *(signed)*	Disk imaging	S-000316, Y (broken by S. Iyer at 09:45 to image; documented)	N/A	IT Security forensic workroom
4	2026-10-15 16:10 IST	S. Iyer *(signed)*	A. Desai *(signed)*	Return after imaging	S-000341 (new), Y	Y (E6 verified 12:30)	IT Security locker L-3

E. Hash record

#	Item / file hashed	Algorithm	Hash value	Computed by	Date and time (TZ)	Tool and version
1	E4 memory image of E1	SHA-256	5664…bfd3b (full value in Section 7)	S. Iyer	2026-10-14 12:15 IST	Examiner's validated toolset (version in examiner notes)
2	E6 disk image of E1, acquisition	SHA-256	9ba1…7611e	S. Iyer	2026-10-15 11:02 IST	As above
3	E6 working copy, verification	SHA-256	9ba1…7611e (match)	S. Iyer	2026-10-15 12:30 IST	As above
4	N/A					

F. Final disposition

Field	Entry
Outcome	Retained pending investigation outcome (status as of 2026-10-15)
Details	To be completed at case closure
Authorised by	Pending
Witness	Pending
Date and time	Pending

9. First responder's hand-over summary (R. Kale)

> At 10:38 IST on 14 October 2026 I took control of college laptop MCAC-IT-LT-0412 at Research Lab 2, Desk 4, following a firewall alert. It was on and unlocked, with no visible destruction. The user reported opening an email attachment and closing Word before I arrived. I photographed the scene before touching anything and recorded that the laptop clock was 3 minutes slow. I did not touch the keyboard, mouse, lid or ports. Network isolation was done at the switch by A. Desai. The USB drive was left in place. I handed the powered-on laptop to examiner S. Iyer at 11:42. I did not run any program on the laptop, open any file, or copy anything from it. My photographs and notebook scan are hashed (E5, E7).

10. How this example follows the guide

Guide step	What happened here	Where to see it
1. Secure the scene	User moved away; reported actions recorded separately	Log 10:38–10:39
2. Observe and record	Ten numbered photos; clock offset measured	Sections 4 and 6
3. Decide power state	ON, no destruction, so kept powered; reason recorded	Section 5
4. Isolate network	Network-side (switch port), no contact with laptop	Log 10:45–10:46
5. Document time and actions	Contemporaneous log with time zone and a proper correction	Section 6
6. Protect original	Bagged and sealed after power-off; locker storage	Log 12:20–12:30
7. Hash where appropriate	SHA-256 for images and responder-created files; none on live E1	Section 7
8. Chain of custody	Four signed transfers; seal break and reseal documented	Section 8
9. Hand over and report	Written summary of what was and was not done	Section 9

What would have gone wrong with common mistakes:

- **"Shut down" from the Start menu:** memory evidence (E4) lost, BitLocker locks the disk, and Fast Startup writes hibernation data to disk.
- **Pulling out the USB drive to "check it" on another PC:** that PC could change the drive's contents or be infected itself. The drive's original position would also be undocumented.
- **Letting the user "show" what happened:** new activity overwrites the very evidence that shows how the compromise began.
- **No clock offset:** every laptop timestamp would be 3 minutes out of line with firewall logs, which weakens the timeline.